

Mohamed Ali

Senior Machine Learning Engineer

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SUMMARY

Senior Machine Learning Engineer with robust experience in machine learning, Generative AI, data science, and software engineering. Proven ability to deliver production-grade AI solutions in dynamic environments, specializing in sales forecasting, fraud prevention, recommendation systems, and advanced NLP for business impact.

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer, WeBook

06/2024 – Present

Saudi Arabia Remote

- Architected and developed the **WeBook Brain**, an integrated AI system featuring a Gemini 2-powered chatbot, a high-performance Fraud Detection RAG application, and a sophisticated recommendation system.
- Engineered a scalable FastAPI backend for the fraud detection module, using Ollama 3, Mistral, ChromaDB, and MongoDB, and containerized the solution with Docker for cross-platform deployment.
- Developed a robust churn prediction model by leveraging advanced machine learning algorithms and feature engineering to enhance customer retention strategies.

Senior Machine Learning Engineer, IbnSina Pharma

06/2023 – 05/2024

Egypt

- Built and implemented advanced predictive models for sales forecasting, customer segmentation, fraud prevention, and churn prediction to drive strategic business decisions.
- Optimized product demand predictions by deploying ML and DL models, including ARIMA, SARIMA, and Prophet, leading to improved strategic planning.
- Extracted actionable insights from complex datasets to refine marketing campaigns and enhance overall decision-making processes.

AI Engineer, MDS

04/2021 – 05/2023

Saudi Arabia

- Designed, developed, and deployed scalable artificial intelligence systems that utilized large-scale, real-world datasets for industrial research and prediction algorithms.
- Constructed a deep learning model to accurately predict network traffic load for large-scale network devices in a real infrastructure environment and successfully deployed it into production.

AI & Machine Learning Engineer, Linked Gates

08/2019 – 03/2021

Egypt

- Constructed intelligent software systems using the Odoo framework, integrated with AI models to deliver predictive insights for client-specific use cases.
- Developed an image classification system using Convolutional Neural Networks (CNNs) trained on the ImageNet dataset for high-accuracy categorization.
- Engineered a comprehensive restaurant management platform (VOO) with a mobile application to streamline order management, inventory, and customer engagement.

EDUCATION

M.Sc., Computing and Artificial Intelligence Fayoum University, Egypt	2022 – 2025
Advanced Diploma, AI and Data Science Information Technology Institute (ITI), Cairo	2020 – 2021
B.Sc., Computing and Artificial Intelligence Fayoum University, Egypt	2015 – 2019

PROJECTS

Multi-Lingual RAG (2025)

Developed a Retrieval-Augmented Generation (RAG) system for processing and generating content across multiple languages.

Real-Time Fraud Detection (2025)

Implemented a high-throughput, low-latency system for real-time transaction analysis to prevent financial loss.

Bias Detection in AI Models (2025)

Created a framework to identify and mitigate biases in machine learning models, ensuring fairness and ethical AI performance.

WeBook Assistant (2025)

Built a dual-agent AI assistant based on Gemini 2, combining data API and chatbot for seamless support.

Fraud RAG (2025)

Engineered a fraud detection app using Ollama 3, Mistral, ChromaDB, and MongoDB. Delivered scalable FastAPI backend with Docker.

Anxiety Detection (2024)

Designed a predictive model to detect early signs of anxiety using physiological and behavioral data.

ECG Classification (2024)

Developed a deep learning model for the automated classification of electrocardiogram (ECG) signals to identify cardiac arrhythmias.

Mini-RAG (2024)

Developed a lightweight RAG application using Mistral LLM and txtai for semantic search.

Brain Tumor Segmentation (2024)

Developed a YOLOX-based X-ray system for hyper-resolution brain tumor segmentation and classification, enhancing diagnostic accuracy.

Churn Prediction (2024)

Designed an ML pipeline using advanced feature engineering and ML algorithms to improve customer retention.

Smart Robot (2024)

Designed AI model for solar panel cleaning robots, integrating dust sensors for efficiency optimization.

Sales Forecasting (2023)

Implemented ARIMA, SARIMA, and Prophet for best-in-class sales prediction.

Fatigue Prediction (2023)

Created a hybrid learning pipeline (CatBoost, XGBoost, LightGBM, deep NN) for fatigue detection.

Network Traffic Load Prediction (2022–2023)

Built and deployed a DL model to forecast device load in network infrastructure environments.

Arabic OCR (2022)

Developed Arabic text extraction systems using Docker, Flask, and REST API for document digitization.

Video Captioning (2021–2023)

Conducted research using ResNet, VGG, LSTM, and Glove to generate natural captions for video content.

Image Classification System (2020)

Created CNN-based system trained on ImageNet for high-accuracy multi-category classification.

SCADA Innovation (2020)

Automated manufacturing software using Odoo and Postgres for high-precision production.

VOO Platform (2019–2020)

Built restaurant management platform and mobile app for streamlined order and inventory control.

TECHNICAL SKILLS

- **Programming:** Python, R, C++, Bash Scripting
- **ML/DL:** TensorFlow, PyTorch, Scikit-Learn, Hugging Face, OpenCV, LangChain, MLFlow, NumPy
- **GenAI:** LLM Finetuning, Prompt Engineering, AI Agents, RAG
- **Big Data:** PySpark, Pandas, Dask
- **Vector DBs :** ChromaDB, FAISS, Qdrant, Pinecone
- **MLOps:** Docker, Kubernetes, GitHub Actions, AWS SageMaker, AzureML, MLFlow, CI/CD, DVC
- **BI/Visualization:** Power BI, Streamlit, Gradio, Plotly, Matplotlib, Seaborn
- **Frameworks:** FastAPI, Flask, Odoo

SOFT SKILLS

- Problem Solving
- Teamwork
- Leadership

HONORS

- 1st Place, Pre-master Class, Fayoum University (2022)
- 1st Place, Graduation Projects, Fayoum University (2019)
- 3rd Place, IEEE-ICPC Programming Competition (2018)

VOLUNTEER EXPERIENCE

- GDG Fayoum
- IEEE-FSB
- Life Makers

LANGUAGES

- **Arabic:** Native
- **English:** Full Professional